0. Definitions *coordinates, the slope a'_{hd}, and the closed-loop tracking error*

$$h[k] = h_d[k] - \delta h[k], \qquad \Delta h_d[k] = h_d[k+1] - h_d[k]$$

$$a'_{hd}[k] := \left.\frac{da_h}{dh}\right|_{h_d[k]}, \qquad a_r[k] := a_h[k] - a_h\big(h_d[k]\big)$$

$$\delta h[k+1] = \lambda_c\, \delta h[k] - F_{dh}[k]\, e_{a_h}[k] - \varepsilon_h[k]$$

> a_h = actual-height gain (state 4); a'_{hd} = its slope (state 5); a_r = residual gain from the tracking error. $a_{var} = a_{cov} = 0.05$ (EWMA weights); $\beta = 1 - T_s/\tau$ ($\tau = 0.35\,\mathrm{s} \Rightarrow \beta = 0.99821$) = the a' smoothing weight. $\delta h_1 := \delta h[k-d]$.

1. True *the residual is $-a'_{hd}\,\delta h$; hence the var-ratio recovers a'_{hd}, and the β-EWMA is its update law*

First-order Taylor of the gain at the perturbed height:

$$a_h\big(h[k]\big) \approx a_h\big(h_d[k]\big) - a'_{hd}[k]\, \delta h[k]$$

$$\boxed{a_r[k] = -\, a'_{hd}[k]\, \delta h[k], \qquad a'_{hd}[k+1] \approx a'_{hd}[k]}$$

Because the residual and the tracking error share the author a'_{hd}, their variance ratio recovers the slope:

$$a'_{hdm}[k] := +\sqrt{\mathrm{Var}\big(a_r[k]\big) \,/\, \mathrm{Var}\big(\delta h[k]\big)} = a'_{hd}[k] \qquad (a'_{hd} > 0)$$

The true update law for the slope is a β-EWMA of this readout — consistent because $a'_{hdm} \equiv a'_{hd}$:

$$\boxed{a'_{hd}[k+1] = \beta\, a'_{hd}[k] + (1-\beta)\, a'_{hdm}[k], \qquad a'_{hdm} \equiv a'_{hd}}$$

2. $\hat{a}'_{hdm}$ *the estimator's empirical var-ratio: two EWMA chains on $\hat{a}_h$ and $\delta\hat{h}_3$*

Replace the true quantities by their estimates. One chain tracks the mean and variance of the gain estimate, the other of the tracking-error estimate:

$$\overline{a}_h[k+1] = (1 - a_{var})\, \overline{a}_h[k] + a_{var}\, \hat{a}_h[k+1], \qquad \hat{a}_r[k] = \hat{a}_h[k] - \overline{a}_h[k]$$

$$\hat{\sigma}^2_{\hat{a}_r}[k+1] = (1 - a_{cov})\, \hat{\sigma}^2_{\hat{a}_r}[k] + a_{cov}\, \hat{a}_r^2[k+1]$$

$$\overline{\delta\hat{h}}[k+1] = (1 - a_{var})\, \overline{\delta\hat{h}}[k] + a_{var}\, \delta\hat{h}_3[k+1], \quad \delta\hat{h}_r[k] = \delta\hat{h}_3[k] - \overline{\delta\hat{h}}[k]$$

$$\hat{\sigma}^2_{\delta\hat{h}}[k+1] = (1 - a_{cov})\, \hat{\sigma}^2_{\delta\hat{h}}[k] + a_{cov}\, \delta\hat{h}_r^2[k+1]$$

$$\boxed{\hat{a}'_{hdm}[k] := +\sqrt{\hat{\sigma}^2_{\hat{a}_r}[k] \,/\, \hat{\sigma}^2_{\delta\hat{h}}[k]}}$$

3. Esti (co-update) *a'_{hd} is updated by the β-EWMA of $\hat{a}'_{hdm}$* plus *the KF innovation correction*

State $x = [\,\delta h_1,\ \delta h_2,\ \delta h_3,\ a_h,\ a'_{hd}\,]^\mathsf{T}$. The KF proper corrects $[\delta h_1, \delta h_2, \delta h_3, a_h]$ from $y_1 = \delta x_m$ and $y_2 = a_{xm}$; innovations $e_{y_1} = y_1 - \hat{y}_1$, $e_{y_2} = y_2 - \hat{y}_2$; gain $K = P^- H^\mathsf{T} S^{-1}$, $S = HP^-H^\mathsf{T} + R$.

$$\hat{a}_h[k+1] = \hat{a}_h[k] + \hat{a}'_{hd}[k]\big(\Delta h_d[k] + (1-\lambda_c)\,\delta\hat{h}_3[k]\big) \qquad \text{(gain-level predict)}$$

$$\boxed{\hat{a}'_{hd}[k+1] = \underbrace{\beta\,\hat{a}'_{hd}[k] + (1-\beta)\,\hat{a}'_{hdm}[k]}_{\text{(i) var-ratio }\beta\text{-EWMA (§1 form)}} \;+\; \underbrace{L_{51}[k]\,e_{y_1}[k] + L_{52}[k]\,e_{y_2}[k]}_{\text{(ii) KF }L\cdot e\text{ correction}}}$$

$L_{5j} = [P^- H^\mathsf{T} S^{-1}]_{5,j}$. With $H(\cdot,5) = 0$ (no own measurement row for a'_{hd}), L_{5j} is a *cross-covariance* gain $\propto P^-_{5,4}\,H(j,4)$, built by the coupling $F_e(4,5) = \Delta h_d$. Setting $L{\cdot}e = 0$ recovers the pure esti bolt-on; dropping $(1-\beta)\hat{a}'_{hdm}$ recovers the pure as-state.

4. Error *how the estimate errors evolve*

$e_{a_h} = a_h - \hat{a}_h,\ \ e_{a'_{hd}} = a'_{hd} - \hat{a}'_{hd},\ \ e_h = \delta h - \delta\hat{h}.$

$$e_{a_h}[k+1] = \big(1 + \hat{a}'_{hd}F_{dh}\big)\,e_{a_h} + \Delta h_d\,e_{a'_{hd}} + (1-\lambda_c)\,\hat{a}'_{hd}\,e_h + \hat{a}'_{hd}\,\varepsilon_h$$

$$\boxed{e_{a'_{hd}}[k+1] = \beta\,e_{a'_{hd}}[k] \;-\; (1-\beta)\big(\hat{a}'_{hdm}[k] - a'_{hdm}[k]\big) \;-\; \text{(KF correction)}}$$

The middle term is the var-ratio readout error (self-copy + noise); the last is the genuine measurement pull — the $L{\cdot}e$ term of §3, kept symbolic here.

5. $\mathbf{F}_e$ and H *the 5×5 transition and 2×5 measurement Jacobians*

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh}[k] & 0 \\ 0 & 0 & (1-\lambda_c)\hat{a}'_{hd}[k] & 1 + \hat{a}'_{hd}[k]F_{dh}[k] & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & \beta \end{bmatrix}, \qquad H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Row-5 pole $= \beta$ (the EWMA memory); the $(1-\beta)\hat{a}'_{hdm}$ drive and the $L{\cdot}e$ correction enter as inputs, not through the pole. $F_e(4,5) = \Delta h_d$ is the *only* path giving a'_{hd} a Kalman gain.

6. Q and R *process and measurement noise*

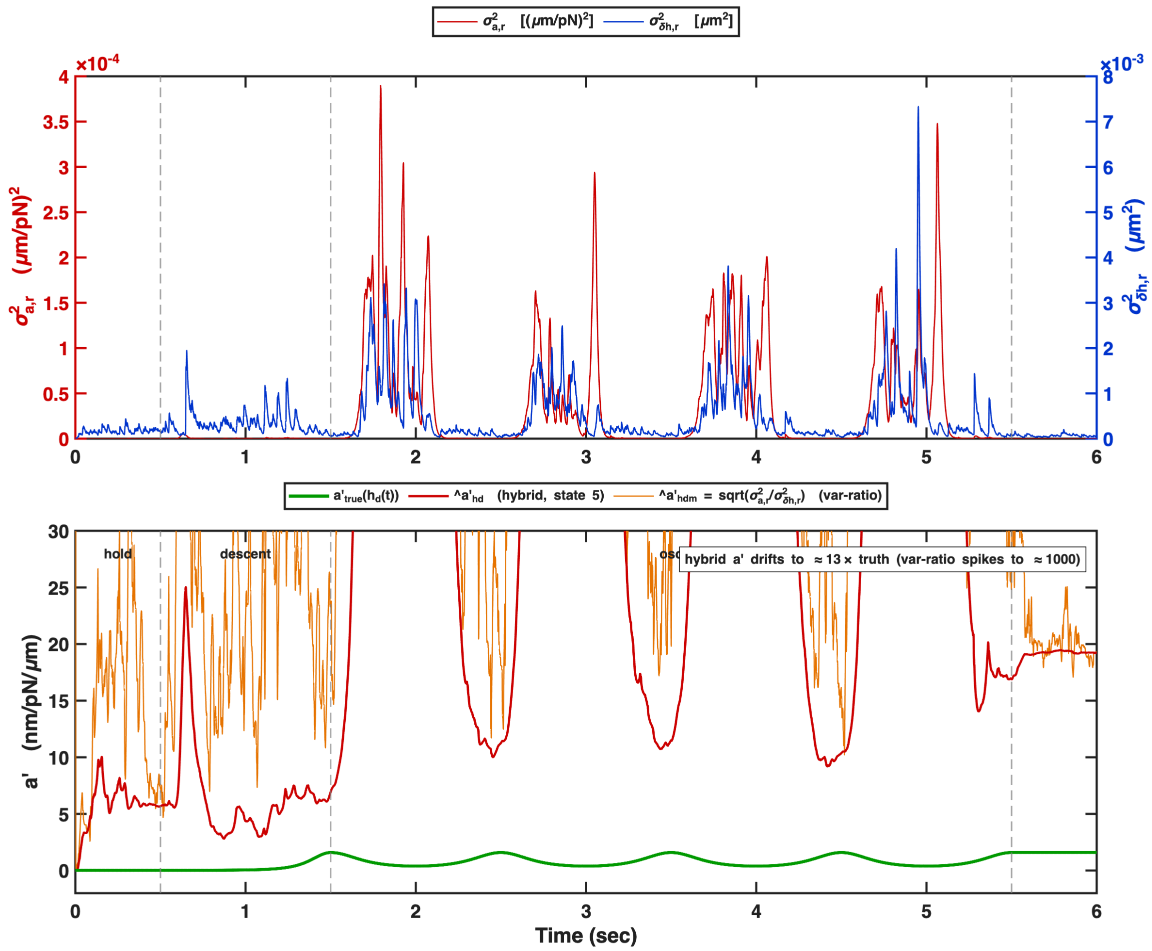
$$Q = \mathrm{diag}\big(0,\ 0,\ Q_{33},\ Q_{44},\ Q_{55}\big), \qquad R = \mathrm{diag}\big(\sigma_n^2,\ R_2\big)$$

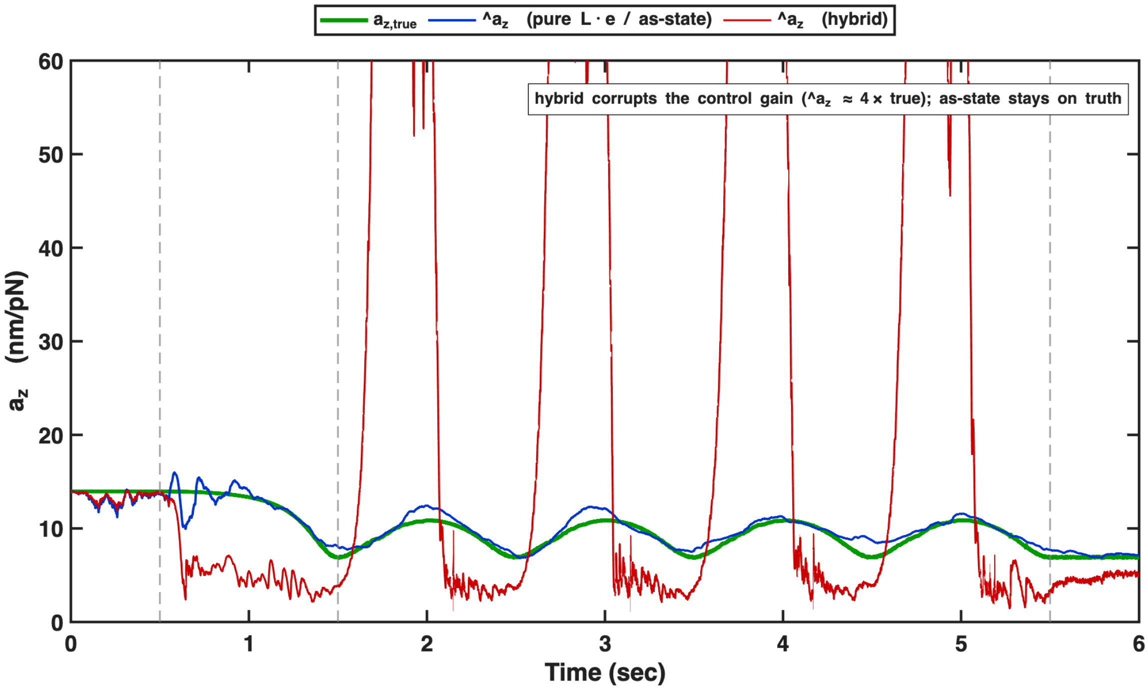
$$Q_{33} = 4k_BT\,\hat{a}_h\big(1 + (1-\lambda_c)^2 d\big) + (1-\lambda_c)^2\sigma_n^2, \qquad Q_{44} = \big(\hat{a}_h K_h/R\big)^2\,\sigma_{\delta h}^2, \qquad Q_{55} = (1-\beta)^2\,R_{var}$$

$$R_2 = a_{cov}\,IF_{var}\,\big(\hat{a}_h + \xi\big)^2, \qquad R_{var} \approx \tfrac{a_{cov}}{4}\big(IF_{\hat{a}_r} + IF_{\delta\hat{h}} - 2IF_{\times}\big)\,\hat{a}_{hd}'^{\,2}$$

$\sigma_{\delta h}^2 = 4k_BT\,\hat{a}_h$; $\xi = (C_n/C_{dpmr})\,\sigma_n^2/(4k_BT)$; $IF_* = 1 + 2\sum_{j\geq 1}\rho_*(j)(1-a_{cov})^j$. Q_{55} is the injected variance of the $(1-\beta)\hat{a}_{hdm}'$ drive.

7. Simulation (1 Hz) *co-update traces, init* $1\times$





8. As-state (the blue line): drop the var-ratio

a'_{hd} is a plain KF state, corrected by $L\cdot e$ only — no $\hat{a}'_{hdm}$

Start from the same True relation (§1): $a_r[k] = -a'_{hd}[k]\,\delta h[k]$, $a'_{hd}[k+1] \approx a'_{hd}[k]$. The as-state estimator keeps a'_{hd} as a plain 5th KF state and *removes the var-ratio channel entirely* — no $\hat{a}'_{hdm}$, no β-EWMA. State $x = [\,\delta h_1, \delta h_2, \delta h_3, a_h, a'_{hd}\,]^\mathsf{T}$.

$$\text{Predict:}\quad \hat{a}_h[k+1] = \hat{a}_h[k] + \hat{a}'_{hd}\big(\Delta h_d + (1-\lambda_c)\,\delta\hat{h}_3\big), \qquad \hat{a}'_{hd}[k+1] = \hat{a}'_{hd}[k] \quad \text{(integrator)}.$$

$$\boxed{\text{Update (KF only):}\qquad \hat{a}'_{hd}[k+1] = \hat{a}'_{hd}[k] + L_{51}\,e_{y_1} + L_{52}\,e_{y_2}}$$

With $H = \left[\begin{smallmatrix}1&0&0&0&0\\0&0&0&1&0\end{smallmatrix}\right]$ $(H(\cdot,5)=0)$, L_{5j} is a *cross-covariance* gain $\propto P^-_{5,4}$, built solely by $F_e(4,5) = \Delta h_d$:

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh} & 0 \\ 0 & 0 & (1-\lambda_c)\hat{a}'_{hd} & 1+\hat{a}'_{hd}F_{dh} & \Delta h_d \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad Q_{55} = \sigma^2_{a''}\big(\Delta h_d^2 + \sigma^2_{\delta h}\big).$$

Row-5 pole $= 1$ (pure integrator, cf. β in the co-update); a'_{hd} moves *only* through the Le correction. $\sigma^2_{a''} \sim (a_{nom}/R^2)^2$ is the sole curvature prior. This is the blue line: $\hat{a}_z$ on truth (1.04×), tracking 26.5 nm; a'_{hd} underestimated in holds ($\Delta h_d \to 0$, cross-cov gain $\to 0$ — the hold-desert CRLB floor).

8.1 $\hat{a}_z$ jitter, and its first-hold amplification when a'_{hd} is unobservable *a leak-free gain integrator; the sole measurement a_{xm} sets a floor that an unobservable a'_{hd} exceeds*

Gain: a leak-free integrator, one noisy measurement, two $\hat{a}'_{hd}$-driven inputs:

$F_e(4,4) \approx 1, \qquad$ measurement $y_2 = a_{xm}$ (rel. std 0.37–0.54),

$$\hat{a}_z[k{+}1] = \hat{a}_z[k] + \underbrace{(1{-}\lambda_c)\,\hat{a}'_{hd}\,\delta\hat{h}_3[k]}_{\text{(B) predict injection}} + \cdots, \qquad \underbrace{Q_{44} = \hat{a}'^{\,2}_{hd} Q_{33}}_{\text{(Q) process noise}}.$$

a'_{hd} enters y_2 *only* through the gain, with regressor φ; hence its Fisher information and CRLB:

$$\varphi[k] := \Delta h_d[k] + (1-\lambda_c)\delta\hat{h}_3[k], \qquad e_{a_h}[k{+}1] \supset \varphi[k]\, e_{a'_{hd}}[k],$$

$$\mathcal{I}(a'_{hd}) = \frac{1}{S_2}\sum_k \varphi^2[k], \qquad \mathrm{Var}(\hat{a}'_{hd}) \geq \mathcal{I}^{-1}(a'_{hd}) \qquad \text{(CRLB)}.$$

$$\boxed{\text{far hold: } \Delta h_d = 0 \Rightarrow \varphi = (1{-}\lambda_c)\delta\hat{h}_3 \Rightarrow \mathcal{I}(a'_{hd}) \to 0 \Rightarrow \hat{a}'_{hd} \text{ drifts } (\hat{a}'_{hd}\to 6\text{–}13 \text{ vs } a'_{hd}\approx 0)}$$

The drifting $\hat{a}'_{hd}$ drives both inputs; the gain jitter scales with them:

$$\text{(B): } \sigma_{\text{inj}} = (1{-}\lambda_c)\,|e_{a'_{hd}}|\,\sigma_{\delta h}\ \text{/step}; \qquad \text{(Q): } P_{44} \propto Q_{44} \propto \hat{a}'^{\,2}_{hd} \Rightarrow L_4 \propto P_{44} \Rightarrow \mathrm{Var}(\hat{a}_z)|_{\text{meas}} \sim L_4^2 R_2.$$

$$\boxed{\hat{a}'_{hd} \to 0 \Rightarrow \text{(B),(Q)} \to 0 \Rightarrow \mathrm{std}(\hat{a}_z) = 0.015; \qquad \hat{a}'_{hd} \text{ drifting} \Rightarrow \mathrm{std}(\hat{a}_z) = 0.61\ (40\times).}$$

The $\sim$5% far-wall hold jitter ($\langle|\hat{a}_z - a_z|\rangle = 0.73$ nm/pN) is thus the *removable* $\hat{a}'_{hd}$-drift, not the a_{xm} floor (0.015, frozen $\hat{a}'_{hd}$). First-hold fix: disable the $\hat{a}'_{hd}$-update where $\mathcal{I}(a'_{hd}) \to 0$ (freeze during the hold, resume at the first motion); accurate-$\hat{a}_z$ levers in §8.2.

8.2 Two charter-legal fixes for accurate $\hat{a}_z$ *(a) a reverting a_d-state removes the drift; (b) an independent c-free channel C_2 passes the a_{xm} floor*

Charter: $c(\bar{h})$ unknown; only a_{nom}, R, T_s, wall position, commanded trajectory usable.

(a) a_d-state (`kfmeas`): reversion, not a free integrator.

$a_d := a_h(h_d)$ (desired-height gain), $\qquad a_d[k{+}1] = a_d[k] + \hat{a}'_{hd}\,\Delta h_d[k]$ (deterministic, exogenous).

$$\boxed{\hat{a}_z = a_d - \hat{a}'_{hd}\,\delta\hat{h}_3; \qquad \text{predict-side drift } 2.74\% \to 0.09\%\ (\sim 30\times),\ \text{no } c(\bar{h}).}$$

(b) C_2: a second gain channel, independent of a_{xm}. A gain error produces a *systematic* tracking residual:

$$f_d = \tfrac{1}{\hat{a}}\{\cdots\} \;\Rightarrow\; \delta x_{md} \;\propto\; \frac{a-\hat{a}}{\hat{a}}, \qquad y_3 := \delta x_{md} \text{ (deterministic residual, } = p_{md}).$$

$$\boxed{\mathbb{E}[y_3] = \kappa_{C_2}\,(a-\hat{a}), \quad \kappa_{C_2} \approx 0.90\text{–}0.94 \text{ (unbiased, } c\text{-free)}, \quad \text{floor} \sim 9.4\%\, a/0.5\text{s}.}$$

C_2 measures the gain error directly, independent of a_{xm}; $a_{cov}/Q_{44}/R_2$ tuning only trades jitter for lag and cannot pass the a_{xm}-only floor.

9. Appendix: the self-copy $\hat{a}'_{hdm} \equiv \hat{a}'_{hd}$ ($\kappa \approx 0$) *why the var-ratio carries no truth*

In a hold ($\Delta h_d = 0$) the gain predict reduces to $\hat{a}_h[k+1] = \hat{a}_h[k] + (1-\lambda_c)\,\hat{a}'_{hd}\,\delta\hat{h}_3[k]$. Using the telescoping identity $(1-\lambda_c)\,\delta\hat{h}_3[j] = \delta\hat{h}_3[j] - \delta\hat{h}_3[j+1]$ and summing from k_0,

$$\hat{a}_h[k] = \hat{a}_h[k_0] + \hat{a}'_{hd}\big(\delta\hat{h}_3[k_0] - \delta\hat{h}_3[k]\big) \quad\Longrightarrow\quad \hat{a}_r[k] = -\,\hat{a}'_{hd}[k]\,\delta\hat{h}_r[k].$$

So $\hat{a}_r$ is built *from* $\hat{a}'_{hd}$, and the two variances are locked:

$$\hat{\sigma}^2_{\hat{a}_r} = {\hat{a}'_{hd}}^2\,\hat{\sigma}^2_{\delta\hat{h}}$$

$$\boxed{\hat{a}'_{hdm} = \sqrt{\hat{\sigma}^2_{\hat{a}_r}/\hat{\sigma}^2_{\delta\hat{h}}} = \hat{a}'_{hd} \qquad \text{for } \textit{any} \text{ true } a'_{hd}}$$

The readout echoes the estimate, independent of the truth; its fidelity is

$$\kappa := \frac{\partial\,\hat{a}'_{hdm}}{\partial\,a'_{\text{true}}} = 0.$$

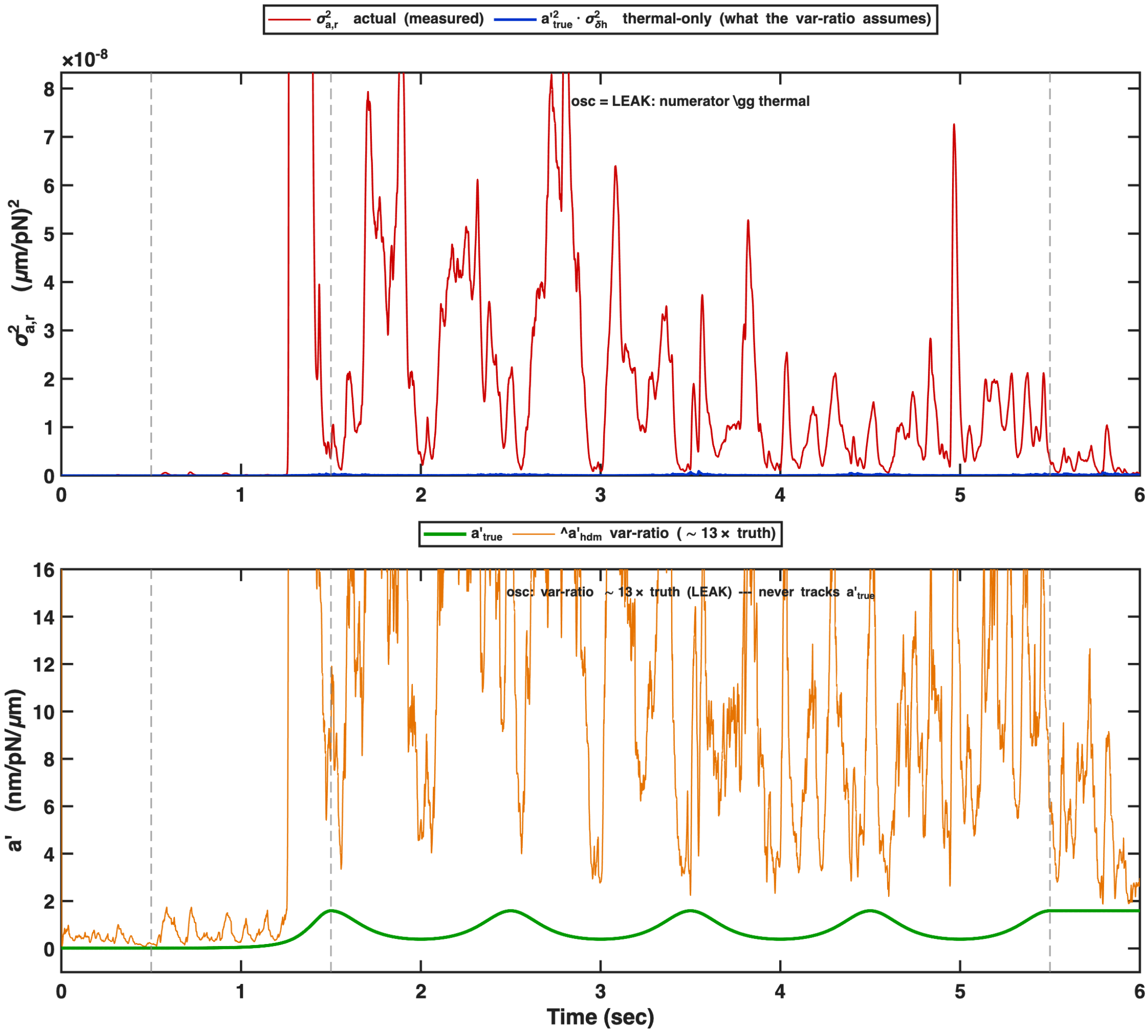
(The measured $\kappa \approx 0.03$ and the +2.5% sqrt-rectification bias $b_u \propto 1/\hat{a}'_{hd}$ come from the correction terms and finite EWMA windows; both are $\ll 1$, so the echo dominates.)

10. Appendix: the leak — the motion face of the self-copy *why the var-ratio is garbage in motion, and why the leak cannot be fixed*

In motion the gain estimate $\hat{a}_h$ swings to track the changing desired-height gain $a_{\text{det}}(t)$ (via the feedforward $\hat{a}'_{hd}\,\Delta h_d$). The mean-tracker $\overline{a}_h = \text{EWMA}(\hat{a}_h)$ is a low-pass; its complementary high-pass $\hat{a}_r = \hat{a}_h - \overline{a}_h$ does *not* remove this swing fully — it leaks a fraction

$$\lambda_{\text{leak}} \approx \frac{2\pi f\,T_s}{a_{var}} \approx 0.079$$

of it into $\hat{a}_r$. Since the deterministic swing (~nm/pN) dwarfs the thermal residual $a_r = -a'_{hd}\,\delta h$ ($\sim$ 0.04 nm/pN), even a 7.9% leak dominates: $\sigma^2_{\hat{a}_r}$ inflates far above the thermal ${\hat{a}'_{hd}}^2\sigma^2_{\delta\hat{h}}$, so $\hat{a}'_{hdm} = \sqrt{\sigma^2_{\hat{a}_r}/\sigma^2_{\delta\hat{h}}} \gg \hat{a}'_{hd}$.



Diagnostic var-ratio computed from a healthy as-state $\hat{a}'_{hd}$. Top: the actual numerator $\sigma^2_{a,r}$ (red) sits $\sim 170\times$ above the thermal-only value it should have (blue) during osc — the leak. Bottom: the var-ratio $\hat{a}'_{hdm}$ (orange) is $\sim 13\times$ truth (green) — garbage — during motion, and never tracks a'_{true}.

Why it cannot be fixed. The leaked swing is itself $\hat{a}'_{hd}$-driven (the feedforward uses $\hat{a}'_{hd}$), so $\hat{a}'_{hdm}$ stays $\hat{a}'_{hd}$-derived — the leak is the *motion* face of the same self-copy whose *hold* face is §9's echo. Removing the leak (faster mean-tracker, or subtracting the feedforward) only turns motion-garbage back into the hold echo $\hat{a}'_{hdm} \equiv \hat{a}'_{hd}$ — still no truth; the root ($\hat{a}_r$ built from $\hat{a}'_{hd}$) is untouched. The clean fix is not to repair the var-ratio but to drop it (pure $L{\cdot}e$ / as-state, or `5state_aprime_kf_meas`).